

Cybersecurity for Connected Cars: Edge-Based Federated Intrusion Detection

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Supervisor :

A/Prof Ke (Desmond) Wang

Students :

Likhith Lokesh Gowda (s4062973)

Verna Nakhla (s3945172)

Joshua Wong (s3944445)

Ken Navarro (s4005415)

Andrew Ng (s4004645)

Research Mentor :

Mr. Kanwardeep Singh Gahlot



Assessment Task 1

Project Proposal and Risk Assessment

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1. Executive Summary

Connected vehicles broadcast safety messages ten times a second to nearby vehicles, road infrastructure, and cellular networks. These messages carry position, speed, and brake data that other vehicles use to make split-second decisions. The protocol that handles this deliberately skips authentication to stay within the required timing window, which means every message is unauthenticated by design. Attacks like position spoofing, replay injection, and denial-of-service flooding are straightforward to execute as a result. Regulations including UNECE No. 155 and ISO/SAE 21434 now require manufacturers to actively manage these risks, and automotive cybersecurity has become a legal obligation in regulated markets.

Detecting these attacks quickly enough is where existing systems fall short. Direct vehicle-to-vehicle communication has a 100-millisecond timing window, and cloud-based detection adds 200 to 400 milliseconds of delay on top. By the time an alert arrives, the vehicle has already acted on the malicious message. Moving detection to roadside edge nodes resolves the timing issue but creates two new ones: streaming raw vehicle traffic is not practical at urban scale, and the GPS coordinates and movement histories in each message are personal data under GDPR, making centralised collection a legal risk.

Federated learning addresses all three. Each roadside node trains a local detection model on its own traffic and shares only the weight updates with a central aggregation server, which combines them into a shared global model. No raw vehicle data leaves the device. This approach has already demonstrated 95% bandwidth reduction with accuracy matching centralised systems, but it has only been tested on generic network data. No published study has evaluated a federated IDS on the vehicular misbehavior data or 5G network-layer attack traffic that C-V2X environments actually produce. That is the gap this project addresses.

The project runs across two semesters. Part A, covered in this proposal, builds the core system: a labelled 5G C-V2X intrusion dataset generated through network simulation, feature engineering on vehicular misbehavior data, a multiclass classification baseline, and a federated learning prototype with initial evaluation results. Part B extends this with full experimental analysis, aggregation robustness testing under varied data conditions, cross-dataset comparison, and an investigation into edge hardware feasibility.

Performance across both parts is evaluated against detection accuracy, per-class F1-score, Matthews Correlation Coefficient, inference latency relative to the 100-millisecond limit, and bandwidth overhead compared to a centralized baseline.

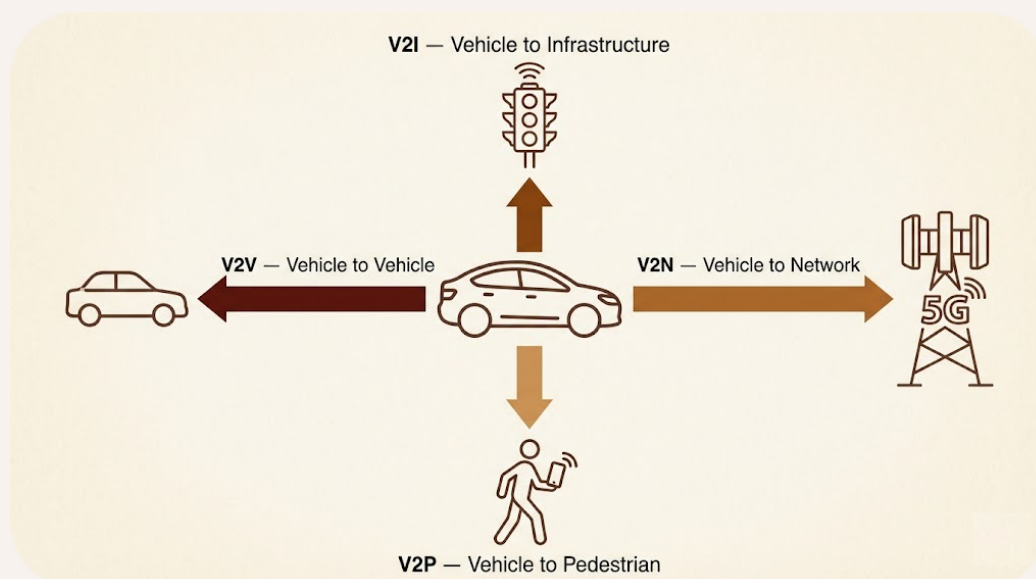


Figure 1 — Overview of Cellular Vehicle-to-Everything (C-V2X) communication channels

2. Statement of Problem

2.1 Context and Significance

Modern vehicles are no longer isolated machines. Through a framework called Cellular Vehicle-to-Everything (C-V2X), they continuously exchange data with nearby vehicles (V2V), road infrastructure such as traffic lights and gantries (V2I), cellular networks (V2N), and pedestrians carrying smart devices (V2P). These four communication channels form the backbone of Intelligent Transport Systems (ITS), enabling time-sensitive applications like collision avoidance, emergency braking coordination, and cooperative lane changing.

Vehicles in a C-V2X network exchange short standardized packets called Basic Safety Messages (BSMs) at a rate of ten per second. Each message carries the sender's position, speed, heading, and brake status. A spoofed BSM reporting a false obstacle can trigger emergency braking across an entire convoy, and a replayed hazard warning can redirect vehicles into oncoming traffic. These are real operational risks, not theoretical edge cases.

The global connected car market was valued at approximately USD 65.2 billion in 2024 and is forecast to reach USD 268.5 billion by 2032 [1]. Alongside this growth, remotely executed attacks now account for the majority of recorded automotive cyber incidents, with large-scale events targeting millions of vehicles at once becoming increasingly common [18]. Regulatory bodies have responded with binding requirements. UNECE Regulation No. 155, effective July 2022, requires manufacturers to maintain a Cybersecurity Management System covering 69 catalogued attack vectors [2]. ISO/SAE 21434:2021 mandates cybersecurity risk assessment at every stage of vehicle engineering [3]. Real-time threat detection is now a legal requirement in regulated markets, not a design preference.

2.2 The Engineering Problem

Securing C-V2X networks is harder than securing conventional enterprise networks because three constraints operate simultaneously.

- ♦ **Latency.** The PC5 sidelink interface used for direct vehicle-to-vehicle communication has a maximum end-to-end latency limit of 100 ms, as defined in 3GPP TS 22.185 [4]. A cloud-based detection system adds 200 to 400 ms of round-trip delay [5], meaning a detection result arrives after the vehicle has already acted on the malicious message.
- ♦ **Bandwidth.** In dense urban deployments, the PC5 channel must handle BSM traffic from a large number of vehicles broadcasting simultaneously at 10 Hz, creating substantial processing demands even before any intrusion analysis takes place [4][5]. Streaming this raw traffic to a central server is not feasible at that scale.
- ♦ **Privacy.** Raw V2X traffic contains GPS coordinates, velocity data, and timestamped movement histories. Under GDPR (Regulation (EU) 2016/679), this constitutes personal data subject to strict processing limits [7], and centralizing it creates a high-value target for adversaries.

These three constraints reinforce each other. A system that processes data locally to meet the latency limit still needs to share threat intelligence across nodes without centralizing raw data. That tension is the core engineering problem this project addresses.

2.3 Why Existing Approaches Fall Short

Machine learning-based IDS research has produced strong accuracy results, but almost entirely on centralized systems that do not account for deployment latency [8]. Edge computing moves detection closer to the vehicle, but raw data still travels from the vehicle to the roadside node, leaving bandwidth and privacy unresolved [5]. An isolated edge node also has no visibility into coordinated attacks that span multiple parts of the network.

Federated Learning (FL) resolves all three constraints by having each node train locally and send only model weight updates to a central server, rather than raw traffic. In a simulated enterprise network, this approach achieved up to 95% bandwidth reduction while maintaining detection accuracy comparable to centralized approaches [5]. However, no existing work has validated a federated IDS on vehicular datasets that include the position, timing, and kinematic data specific to C-V2X attacks. Section 3 reviews the existing literature to identify the specific shortcomings that motivate each design decision.

2.4 Project Scope and Proposed Approach

This project designs, simulates, and evaluates an edge-based federated learning IDS for C-V2X networks. Detection operates at the roadside unit (RSU) or gNodeB layer, with each node training a local model and sending only weight updates to a central aggregation server. The system architecture is described in Section 5. The project is structured across three scope tiers.

- ♦ **Core MVP:** A new 5G C-V2X intrusion dataset generated through network simulation, combined with feature engineering on vehicular misbehavior data, a multiclass classification baseline, and a working FedAvg FL prototype with initial evaluation results.
- ♦ **Supporting Goals:** Aggregation that handles differences in data distribution across nodes, and a cross-dataset comparison with a standard network security benchmark. These workstreams begin in Part A and are completed in Part B.
- ♦ **Stretch Goals:** Byzantine-resilient aggregation and edge hardware feasibility testing within the 100 ms latency constraint. Both are planned as Part B activities.

Success will be measured across detection accuracy, F1-score, Matthews Correlation Coefficient (MCC), inference latency, bandwidth reduction relative to a centralized baseline, and convergence behavior under varied data distribution conditions.

This project is delivered across two semesters under the RMIT Engineering Capstone framework. The table below shows the committed scope for each part.

Scope Tier	Part A (This Semester)	Part B (Next Semester)
Core	System design, 5G C-V2X dataset generation (NS-3/Open5GS), VeReMi feature engineering, multiclass classification baseline, and FedAvg FL prototype with preliminary evaluation.	Full experimental analysis, FedProx integration and non-IID robustness testing, centralized vs. federated comparison, and final reporting.
Supporting	UNSW-NB15 preprocessing and feature alignment with VeReMi, if time allows.	Full UNSW-NB15 cross-dataset comparison and validated FedProx results.
Stretch	Simulation-based latency profiling across varying node counts, if core goals are stable.	FPGA edge hardware deployment and Byzantine resilience testing with Krum.

Part A focuses on building the system: generating the dataset, engineering features, implementing the classification baseline, and producing a working FL prototype with initial results. Part B focuses on extending and validating it: completing the FedProx comparison, running full experiments across datasets, and assessing hardware feasibility.

3. Background and Literature Review

This section reviews existing research across four themes: the C-V2X threat landscape, machine learning approaches to intrusion detection, edge computing and federated learning for vehicular IDS, and hardware feasibility at the network edge. The goal is to identify patterns, contradictions, and gaps that inform the project's design choices.

3.1 The Threat Landscape in C-V2X Networks

Attacks on connected vehicles fall into two broad categories. Intra-vehicle attacks target systems inside the car, such as injecting malicious messages on the Controller Area Network (CAN) bus or exploiting the OBD-II diagnostic port to alter engine behavior. Inter-vehicle attacks target the communication channels themselves, and that is where this project focuses. Hakim et al. survey the attack surface across V2V, V2I, V2N, and V2P interfaces, drawing on the 69 threat vectors defined in UNECE Regulation No. 155 [2][6]. The root vulnerability is a deliberate protocol trade-off: the PC5 sidelink skips per-packet authentication to meet the 100 ms BSM timing requirement, so every V2X message is unauthenticated by design [4]. The five attack types most relevant to this project are position spoofing (broadcasting false GPS coordinates), replay attacks (retransmitting captured BSMs out of context), false data injection (corrupting shared driving data), DoS and DDoS flooding (saturating the PC5 channel), and Sybil attacks (one node impersonating many to manipulate collective traffic decisions).

Existing detection results look promising in controlled settings. Random Forest classifiers have reached approximately 92% accuracy on Sybil detection in V2X scenarios, and autoencoder models achieved 94.6% detection rates for Sybil assaults in V2V communication [6]. A 2025 study presented at the IEEE Vehicular Technology Conference, however, showed that adversarial perturbations applied to V2X ML-based IDS models reduced detection accuracy by up to 40% using Fast Gradient Sign Method and Projected Gradient Descent techniques [15]. This finding directly contradicts the robustness implied by the high clean-dataset accuracy figures reported in earlier work, and it represents a genuine conflict in the literature: models that appear highly effective under standard evaluation conditions can be rendered largely ineffective by an attacker who knows the model exists. Clean-dataset accuracy figures therefore significantly overstate real-world performance when an adversary is actively working to evade detection.

3.2 Machine Learning for Intrusion Detection

A range of ML architectures has been applied to intrusion detection. Simpler models such as Deep Neural Networks (DNNs) and Convolutional Neural Networks (CNNs) offer fast inference and work well on structured traffic features. More complex models such as Long Short-Term Memory networks (LSTMs) and Deep Q-Networks (DQNs) can capture temporal patterns and adapt to new attack behavior, but at significantly higher computational cost.

Alkasassbeh et al. test the DQN approach under a strict zero-day evaluation protocol, withholding entire attack families from training and reserving them only for testing [8]. The model achieved 95.9% accuracy on these completely unseen attack types, with a reward function that penalizes missed detections more heavily than false alarms. This asymmetric penalty design suits safety-critical systems and represents one of the strongest generalization results in the current IDS literature.

The broader literature presents a conflict that has not been resolved. Alkasassbeh et al. argue that model complexity is necessary for robust generalization against zero-day attacks [8]. Work on practical vehicular IDS deployment, including Selamnia et al. [5], takes the opposing view: their system uses a simple five-layer DNN precisely because edge nodes cannot sustain the memory and processing demands of RL-based architectures during inference. Both positions are supported by their respective experimental results, and the field has not yet

produced a study that directly measures the detection gain from added model complexity against the latency cost it introduces on constrained edge hardware.

A separate problem runs through most of this literature. Karthiga et al. show that centralized deep learning IDS designs for connected vehicles consistently report accuracy as the primary metric, with no latency measurement included [12]. Karthiga et al. themselves reported a detection latency of 1.75 seconds for their deep learning vehicular IDS model [12], more than seventeen times the 100 ms PC5 safety window. By the time such a system raises an alert, the vehicle has already acted on the malicious message. The gap between reported detection accuracy and timing feasibility is the key unresolved issue across this part of the literature.

3.3 Edge Computing and Federated Learning for Vehicular IDS

Moving detection from the cloud to roadside edge nodes directly addresses the latency problem. An edge-deployed AI-based IDS can process and classify V2X traffic at the RSU, removing the cloud round-trip delay that makes centralized detection infeasible [13]. The bandwidth and privacy problems, however, remain. Raw data still travels from the vehicle to the edge node, and an isolated node has no visibility into attack patterns that span multiple locations across the network.

Federated Learning resolves all three problems at once. Each node trains a local model on its own data and sends only the model weight updates to a central server. The server combines these using the Federated Averaging (FedAvg) algorithm and sends the improved global model back to all nodes. No raw data leaves the local device at any point. Selamnia et al. apply this architecture to C-V2X intrusion detection and report 95.8% accuracy on the CIC-IDS2018 dataset with a 95% reduction in network traffic compared to a centralized system [5]. The model uses a five-layer DNN with feature selection that reduces 80 input features to 30.

This is the most directly relevant prior work to this project, but it has two significant limitations. First, CIC-IDS2018 is a general enterprise network dataset with no GPS coordinates, BSM timing fields, or vehicle kinematic data, so the 95.8% accuracy figure cannot be treated as evidence of vehicular IDS effectiveness against the attack types this project targets. Van der Heijden et al. created VeReMi to fill this gap, providing labelled vehicular misbehavior data from VANET simulations with full BSM and GPS fields [9], yet no published FL-IDS study has evaluated a federated model on it. Second, neither CIC-IDS2018 nor VeReMi captures 5G network-layer attack traffic. Existing 5G intrusion datasets such as 5G-NIDD [19] were collected in non-vehicular testbed environments and do not include the BSM timing, GPS, or mobility features needed for vehicular misbehavior detection.

FedAvg also assumes that data is distributed similarly across all nodes, which is known as the IID assumption. In a vehicular network this rarely holds, as an RSU at a busy city intersection and one on a rural highway see very different traffic patterns and attack types. Li et al. developed FedProx to address this by adding a regularization term to each node's local training objective, improving accuracy by 22% in high-heterogeneity settings [10]. Preliminary findings in a related vehicular FL context also suggest faster convergence under non-IID conditions [14, preprint], though this has not been confirmed in a peer-reviewed vehicular IDS study.

For scenarios where some nodes may be compromised, Blanchard et al. introduced Krum, an aggregation algorithm with a formal Byzantine fault tolerance guarantee [11]. Xie et al. later showed that Krum can be defeated by inner product manipulation attacks, where malicious updates are crafted to appear geometrically close to legitimate ones while still degrading the global model [16]. This finding directly contradicts the reliability guarantees claimed by Blanchard et al. and represents an unresolved tension in Byzantine-resilient FL research, supporting treating Krum as a stretch goal rather than a core component.

3.4 Hardware Feasibility of Edge Inference

The FL-IDS literature rarely addresses whether trained models can run fast enough on real hardware. Simulation studies report accuracy and bandwidth results but almost never verify whether inference could complete within the 100 ms window on a physical RSU or FPGA device.

Work on intra-vehicle IDS provides a useful reference point. Foudhaili et al. describe an FPGA-based IDS for CAN bus attack detection using a quantized neural network built as a dataflow processor [17]. The system achieved accuracy above 99% for DoS and fuzzing attacks and a 15x reduction in power consumption compared to GPU-based equivalents [17], demonstrating that hardware-accelerated neural network inference at the edge is achievable for IDS workloads. Whether inference can complete within 100 ms on RSU or FPGA hardware for the more demanding V2X FL-IDS case, with multi-feature vectors including GPS coordinates and BSM timing fields across a live federated pipeline, has not been tested.

3.5 Identified Research Gaps

- ♦ **Gap 1a: No FL-IDS has been validated on vehicular misbehavior data.** All comparable FL-IDS studies use general enterprise datasets. None have trained or evaluated a federated IDS on VeReMi or any dataset with BSM-level timing and kinematic features [5][9].
- ♦ **Gap 1b: No labelled dataset covers 5G network-layer attacks in a C-V2X context.** Existing datasets either capture vehicular misbehavior at the application layer (VeReMi) or 5G network traffic in non-vehicular settings (5G-NIDD [19]). No publicly available dataset combines both. This gap directly motivates the network simulation approach used in this project, where NS-3 and Open5GS are used to generate labelled 5G C-V2X traffic under controlled attack conditions.
- ♦ **Gap 2: Non-IID data heterogeneity is unaddressed in vehicular IDS.** FedAvg's IID assumption does not hold in vehicular deployments by design. FedProx addresses this in general FL settings [10] but has not been applied to any vehicular intrusion detection task.
- ♦ **Gap 3: Edge hardware feasibility has not been tested for V2X FL-IDS.** FPGA inference has been validated for CAN bus IDS [17], but no study has examined whether FL-IDS models can meet the 100 ms deadline on RSU or FPGA hardware in the V2X context.
- ♦ **Gap 4: Byzantine resilience is untested under vehicular data diversity.** Krum's fault tolerance guarantees are undermined under adversarial manipulation [16]. In vehicular deployments, a separate problem arises: legitimate nodes in geographically different locations naturally produce structurally different model updates, which Krum may incorrectly reject as malicious [11][16].

3.6 Synthesis and Project Justification

The individual components of an effective vehicular FL-IDS already exist in the literature but have never been combined and tested together in the right context. Selamnia et al. show that FL can satisfy bandwidth and privacy constraints for IDS at the network edge [5]. Li et al. show that FedProx can handle data heterogeneity across FL clients [10]. Van der Heijden et al. provide a dataset with the vehicular features needed to evaluate C-V2X attacks properly [9]. Foudhaili et al. establish a hardware reference point for neural network inference at the edge [17]. The 5G-NIDD dataset demonstrates that realistic 5G network intrusion data can be generated from a functional testbed environment [19]. What is missing is a study that brings these components together and evaluates them on vehicular misbehavior data under realistic, heterogeneous conditions, supported by a dataset that reflects both vehicular application-layer behavior and 5G network-layer attacks.

Gaps 1a and 1b are addressed as core Part A contributions: the FL-IDS prototype is trained and evaluated on VeReMi with FedAvg as the baseline, and a new 5G C-V2X dataset is generated through simulation. Gap 2 begins in Part A with the FedAvg baseline and is completed in Part B through FedProx integration and full non-IID robustness testing. Gap 3 is a supporting Part B investigation, with simulation-based latency profiling started in Part A if core work is stable. Gap 4 is a Part B stretch goal. This distribution ensures the core contributions are completed thoroughly in Part A while leaving meaningful, well-defined work for Part B.

4. Design & Research Questions

The four research questions below are derived from the gaps identified in Section 3.5 and are framed to guide the project across both Part A and Part B. Sub-aspects are labelled by scope tier and semester.

RQ1: 5G C-V2X Dataset Generation

No publicly available dataset captures both vehicular mobility characteristics and 5G network-layer attack traffic in a single C-V2X context (Gap 1b). This question drives the dataset generation workstream in Part A and directly affects the quality of every downstream workstream that depends on it.

What simulation parameters, attack scenarios, and network configurations are required to generate a labelled 5G C-V2X intrusion dataset that accurately represents cellular vehicular attack traffic, and how does this generated dataset compare to existing non-vehicular 5G intrusion datasets in terms of feature distribution and attack coverage?

Sub-aspects:

- ♦ (Core, Part A) Which NS-3 or Open5GS simulation parameters, including node density, attack injection rate, and mobility model, most significantly affect the statistical distribution of the generated traffic, and what criteria should be used to validate that the output is representative of realistic 5G C-V2X network conditions?
- ♦ (Supporting, Part B) How does the generated dataset's feature distribution compare to 5G-NIDD and VeReMi across shared attack categories, and what gaps in attack coverage remain after simulation-based generation?

RQ2: Feature Engineering and Multiclass Classification

No FL-IDS has been trained on vehicular misbehavior data with BSM-level kinematic and timing features (Gap 1a), and the literature has not resolved which feature selection approach best captures C-V2X attack signatures. This question covers the feature engineering and multiclass classification workstreams, both core Part A contributions.

Which feature selection method best identifies the kinematic and timing signatures of C-V2X attack types in VeReMi, and how does the resulting feature subset affect multiclass classification performance across the five attack categories?

Sub-aspects:

- ♦ (Core, Part A) How do ANOVA F-score, mutual information, and recursive feature elimination compare in identifying kinematic and timing features specific to individual C-V2X attack types, and does the optimal feature subset change when the same methods are applied to the generated 5G dataset?
- ♦ (Supporting, Part A/B) How does class imbalance between benign traffic and individual attack categories affect precision-recall trade-offs, and which resampling strategy best preserves realistic attack distributions for safety-critical multiclass classification?
- ♦ (Supporting, Part A/B) Which combination of MCC, F1-score, and false positive rate provides the most reliable picture of classification performance on imbalanced vehicular attack datasets, and how should these metrics be reported to allow fair comparison against centralized baseline results?

RQ3: Federated Aggregation Under Data Heterogeneity

FedAvg's IID assumption does not hold in vehicular networks by design, as RSUs in different geographic locations see fundamentally different traffic and attack distributions (Gap 2). This question begins with the FedAvg baseline in Part A and is answered fully in Part B through FedProx experimentation and Byzantine stretch testing.

How does data heterogeneity across vehicular edge nodes degrade FedAvg convergence and detection accuracy, and to what extent does FedProx recover performance under data distributions representative of real C-V2X deployments?

Sub-aspects:

- ♦ (Core, Part B) At what degree of non-IID data divergence does FedAvg degrade below acceptable detection thresholds, and how does this relate to the distributions observed between simulated urban and rural RSU scenarios?
- ♦ (Stretch, Part B) Does structural variation between nodes in different road environments cause Krum to incorrectly reject authentic model updates as malicious, and how does this interact with deliberate Byzantine poisoning attacks?

RQ4: Real-Time Feasibility and Hardware Constraints

No existing FL-IDS study has verified whether inference can complete within the 100ms PC5 safety window on physical edge hardware (Gap 3), and the relationship between model complexity and deployment latency has not been tested in a vehicular context. This question covers the latency profiling and hardware deployment workstream. Latency profiling in simulation is a supporting Part A activity and physical hardware deployment is a Part B stretch goal.

How does inference latency scale with model complexity and vehicular node count, and what are the binding design constraints for meeting the 100ms PC5 requirement under realistic C-V2X traffic conditions?

Sub-aspects:

- ♦ (Supporting, Part A) How does inference latency scale as simulated vehicular node count increases from the minimum viable prototype configuration, and at what density does the 100ms constraint become the binding design limit rather than classification accuracy?
- ♦ (Stretch, Part B) Can the trained FL-IDS model maintain inference latency below 100ms on FPGA or RSU-class hardware, and how does model quantization or compression affect the accuracy-latency trade-off on constrained edge devices?

5. Methodology

This section describes the engineering approach used to investigate the research questions in Section 4. It covers the minimum viable prototype, project phases, dataset preparation, the federated learning pipeline, evaluation strategy, and the resource and budget plan. The full project schedule, including task breakdown, phase boundaries, milestones, and assessment deadlines, is shown in the Gantt chart in Appendix 2.

5.1 Minimum Viable Prototype

The project uses a tiered scope structure to keep commitments achievable within a capstone semester. The minimum viable prototype (MVP) defines five sequential components that must all be complete before supporting or stretch goals begin:

1. A labelled 5G C-V2X intrusion dataset generated through network simulation, with controlled attack injection across the five attack types identified in Section 3.1.
2. A feature-engineered version of VeReMi, reduced to the most discriminative kinematic and timing features using ANOVA F-score and mutual information.
3. A multiclass classification baseline trained on the engineered features and evaluated per attack class using F1-score and MCC.
4. A working FedAvg federated learning simulation with at least three client nodes, each trained on a geographically partitioned subset of VeReMi.
5. Preliminary evaluation results across accuracy, F1-score, MCC, inference latency, and bandwidth overhead relative to a centralized baseline.

Supporting goals (FedProx integration, UNSW-NB15 cross-dataset comparison) begin only when the MVP is complete with at least two weeks remaining. Stretch goals (FPGA hardware feasibility, Krum Byzantine resilience) are Part B activities.

5.2 Engineering Design Approach

The project runs across three sequential phases. No implementation begins until the corresponding design decision is documented in the shared project workspace.

Phase 1: Design and Scoping (Weeks 1 to 5). Finalize the project proposal, confirm the NS-3 or Open5GS simulation configuration, and agree on the VeReMi feature engineering approach. Assessment Task 1 is submitted at the end of this phase.

Phase 2: Prototype Development (Weeks 5 to 10). Dataset generation runs first, feeding feature engineering, which feeds the classification baseline, which feeds the FL prototype. Each workstream publishes its outputs before the next begins.

Phase 3: Evaluation and Reporting (Weeks 10 to 13). FL results are evaluated against the centralized baseline, latency profiling begins if core work is stable, and findings are written up for Assessment Tasks 2 and 3.

The Gantt chart in Appendix 2 maps all tasks across the thirteen-week semester and is grouped by phases and assignments.

Figure 2 — System Architecture Diagram

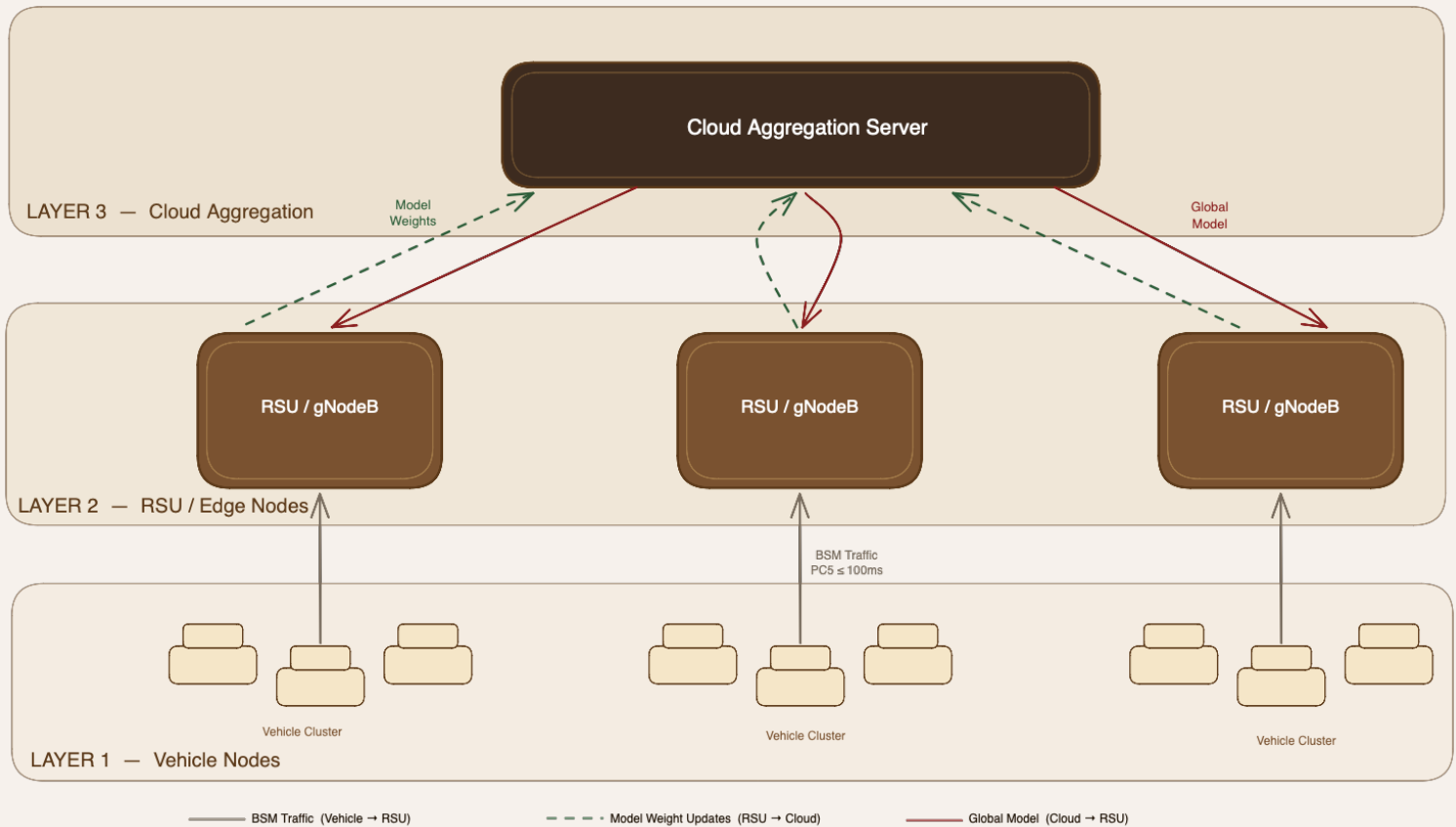
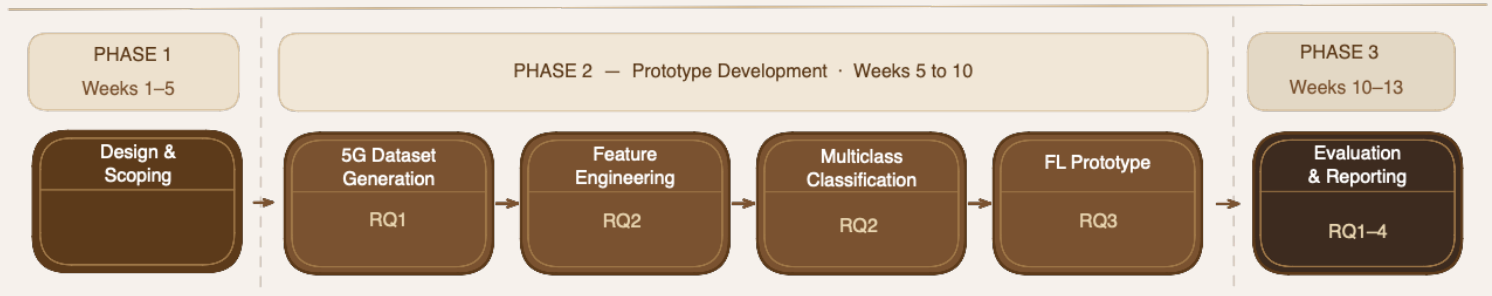


Figure 3 — Methodology Workflow



5.3 Dataset Selection and Preparation

Three datasets serve distinct roles across the pipeline.

Generated 5G C-V2X Dataset. No publicly available dataset combines 5G network-layer attack traffic with vehicular mobility data (Gap 1b). The dataset expansion workstream generates a labelled dataset using NS-3 or Open5GS, with the five C-V2X attack types injected under controlled conditions and the output validated against 5G-NIDD [19].

VeReMi. Primary dataset for feature engineering, classification, and the FL prototype. Provides labelled misbehavior data with full BSM and GPS fields across five attack categories. Publicly available with no licensing restrictions [9].

UNSW-NB15. General network security benchmark for cross-dataset generalization testing. Incorporated as a supporting goal only after the VeReMi pipeline is complete.

All three datasets go through the same preprocessing sequence: duplicate removal, min-max normalization, ANOVA F-score and mutual information feature selection, and SMOTE class balancing. For the FL workstream, VeReMi is split across client nodes using a geographic non-IID partition that reflects the different attack distributions an RSU would observe at different road environment types.

5.4 Federated Learning Pipeline Design

The FL pipeline follows the standard federated loop. The central server distributes the global model; each client trains locally and returns only weight updates; the server aggregates and redistributes. No raw vehicle data leaves any client node at any stage. Three aggregation configurations are evaluated across scope tiers:

Algorithm	Rationale	Scope	Semester
FedAvg	Standard baseline; averages client updates proportionally to dataset size [5].	Core	Part A
FedProx	Proximal regularization stabilizes convergence under non-IID distributions; 22% accuracy gain in heterogeneous settings [10].	Supporting	Part B
Krum	Geometric proximity-based selection for Byzantine fault tolerance [11], but undermined by inner product manipulation attacks [16]. Stretch goal pending FedProx results.	Stretch	Part B

5.5 Simulation Environment and Tools

Tool	Role	Licence
NS-3 / Open5GS	5G C-V2X network simulation and attack injection for dataset generation. Final choice confirmed in Phase 1.	Open-source
MATLAB	ANOVA feature selection and multiclass classification baseline via the Classification Learner Toolbox.	RMIT institutional
Python / Flower / PyTorch	Federated learning pipeline. Flower supports custom aggregation strategies needed for FedProx and Krum in later phases.	Open-source
GitHub	Version control across workstreams with one branch per workstream and mandatory review before merging.	Free

5.6 Evaluation and Analysis Strategy

Performance is measured across accuracy, per-class F1-score, Matthews Correlation Coefficient (MCC), inference latency, and bandwidth overhead relative to a centralized baseline. MCC is the primary aggregate metric because it accounts for all four cells of the confusion matrix and remains meaningful under the class imbalance present in VeReMi, where benign traffic significantly outnumbers individual attack categories. The dataset uses a temporally-aware split so that training data always precedes test data in time, preventing future information from leaking into training.

Three controlled experiments structure the Part A evaluation. The first trains the classification baseline separately on VeReMi, the generated 5G dataset, and UNSW-NB15 to measure how dataset choice affects detection performance for C-V2X attack types. The second evaluates the FL prototype with FedAvg across increasing levels of non-IID data divergence, recording where performance degrades below an acceptable threshold; FedProx

comparison is completed in Part B. The third profiles inference latency as simulated node count scales from the three-node MVP, using the 100ms PC5 safety window as the pass or fail threshold [4]; this begins in Part A if core work is stable and is completed in Part B.

5.7 Alternative Design Approaches

Centralized deep learning only. Cannot address the bandwidth and privacy constraints identified in Section 2.2 and does not scale across multiple RSU nodes. Retained as a comparison baseline only.

Reinforcement learning-based detection. DQN models show strong zero-day generalization [8] but produce higher inference latency than feed-forward networks, conflicting with the 100ms requirement. A candidate for Part B if latency profiling shows the constraint is met with margin.

Standalone edge detection without federation. Resolves the latency problem but loses cross-network threat intelligence. An isolated RSU cannot detect coordinated attacks spanning multiple locations, which is the core limitation federation is designed to overcome.

Full FPGA implementation. Could meet the 100ms constraint with substantial margin, as demonstrated for CAN bus IDS by Foudhaili et al. [17], but hardware procurement and HDL development fall outside the Part A timeline. Defined as a stretch goal for Part B.

5.8 Resource Planning

The key stakeholders are A/Prof Ke (Desmond) Wang as Academic Supervisor, Mr Kanwardeep Singh Gahlot as Industry and Research Mentor (PhD Candidate), and Joshua Wong, Andrew Ng, Verna Nakhla, Likhith Lokesh Gowda, and Ken Navarro as Student Engineers. Desmond sets the research direction and validates design decisions at each weekly meeting. Kanwardeep provides closer technical guidance across workstreams, particularly for dataset generation and FL implementation.

Workstream assignments reflect each team member's specialized area of study:

Workstream	Assigned Lead(s)	Semester
Dataset Expansion (5G generation)	Kanwardeep, Likhith, Verna	Part A
Feature Engineering	Josh, Andrew	Part A
Multiclass Classification	Verna, Ken	Part A
Federated Learning	Likhith	Part A / B
Real-Time Deployment	All members equally	Part B

The workstreams are sequentially dependent. Dataset Expansion feeds Feature Engineering, which feeds Classification, which feeds the FL pipeline. Each workstream publishes its outputs to the shared Notion workspace before the next begins. The workspace uses a central Project Tracker database as the single source of truth for all tasks, milestones, deadlines, and design decisions, with both supervisors having full access. Design decisions are documented in Blueprint pages before implementation begins, following the Blueprint-to-Workbench workflow built into the workspace. All development is distributed across team members' own machines with version control managed through GitHub.

5.9 Budget

All software is open-source or institutionally licensed and all datasets are publicly available. The only anticipated external cost is cloud compute if RMIT's HPC cluster is insufficient during the Phase 2 simulation period. AWS is the nominated fallback, with EC2 for compute and S3 for storage.

Resource	Service	Est. Cost (AUD)	Justification
Cloud compute	AWS EC2 (t3.xlarge)	\$80 to \$150 / month	4 vCPU, 16 GB RAM sufficient for multi-node FL simulation. Cost scales with training rounds and concurrent experiments.
Dataset and checkpoint storage	AWS S3	\$5 to \$15 / month	VeReMi approx. 2 GB, generated 5G dataset approx. 1 to 3 GB, model checkpoints approx. 500 MB.
Primary compute	RMIT HPC Cluster	\$0	Used first. AWS activated only if HPC queue times exceed two working days during peak periods.
Dev tools	NS-3, Open5GS, Python, Flower, PyTorch	\$0	All open-source.
MATLAB	RMIT institutional licence	\$0	Covered under RMIT's existing software agreement.
Datasets	VeReMi, UNSW-NB15, 5G-NIDD	\$0	All publicly available with no licensing fees.
Total estimated cost		\$250 to \$500 AUD	Approx. two to three months of AWS usage in Phases 2 and 3, contingent on HPC availability.

6. Risk & Ethics

6.1 Risk Assessment

A comprehensive risk assessment was carried out for this project, with a risk look-up table matrix in Appendix 1, and the detailed risks provided below. The identified risks were grouped into three categories: project specific risks, system/implementation risks and personal safety risks. Project-specific risks include missed meetings, delays in learning required software tools, task dependencies causing delays in later stages, budget limitations, difficulty obtaining suitable C-V2X or vehicular intrusion datasets, and the possibility of the project becoming too broad to complete within the available time. System and implementation risks include simulated traffic not accurately reflecting real C-V2X behavior, poor IDS performance under changing traffic conditions, simulation becoming too slow at larger scale, and privacy concerns related to shared model updates. Integration complexity is a particularly important risk because incompatibility between components, inconsistent data formats, or difficulties linking the simulator with the IDS and federated learning pipeline may delay development, increase debugging effort and affect the reliability of the final prototype. Personal safety risks are lower because the project is mainly software-based, but basic workspace safety still applies. The sequential dependency of the five workstreams, where Dataset Expansion must complete before Feature Engineering can begin, Feature Engineering before Classification, and Classification before the FL prototype, is also identified as a structural schedule risk and is managed through internal milestones and a defined fallback plan for parallel work.

Strong mitigation strategies have been built into the project plan to lower the risks. To reduce coordination risk in a group setting, regular weekly meetings, meeting minutes, and shared progress tracking are used to improve communication and task alignment. To reduce the broadness of project risk, the project scope is kept focused on a small number of attack types and one clear prototype, while more advanced ideas are treated as optional extensions. Technical risk is reduced by learning tools early, selecting datasets at the start of the project, building the system in stages, and testing each component separately before full integration. Where simulation or training becomes too slow, the team will begin with a smaller setup and scale gradually only if time and computing resources allow. The federated learning workstream is led by a single team member, which is acknowledged as a dependency risk; shared documentation and cross-training on the Flower framework ensure a second team member can take over if needed.

Risk Register

Risk	Likelihood	Impact	Risk Rating	Control Measure	Resultant Risk
Project Specific Risks					
All team members not consistently attending weekly meetings leading to miscommunication and task alignment issues	Likely	Moderate	High	Ensure weekly meetings are held during times available to all team members. Team members make efforts to attend most meetings. Take meeting minutes to share with absent members.	Low
Delays in acquiring training or proficiency with required software tools	Possible	Moderate	High	Allocate time early for learning software tools, use tutorials and supervisor guidance.	Medium
Sequential workstream dependency: Dataset Expansion delays block Feature Engineering, Classification, and the FL prototype from starting on schedule	Possible	Moderate	High	Set a hard internal deadline for Dataset Expansion output by end of Week 7. If missed, begin Feature Engineering on VeReMi alone as a fallback while generation continues in parallel.	Medium
Project costs exceed available budget	Unlikely	Moderate	Medium	Use open-source tools and RMIT software licenses. Activate AWS only if RMIT HPC queue exceeds two working days. Monitor AWS billing with alerts at 50% and 90% of budget.	Low
Difficulty obtaining suitable C-V2X or vehicular intrusion datasets or realistic simulation data	Likely	Moderate	High	Identify public datasets early and confirm suitability with supervisor. Use VeReMi as primary with NS-3/OpenSGS generated dataset as supplement.	Low
The project may become too broad and hard to finish on time	Likely	Severe	Catastrophic	Keep the final scope focused on a small number of attacks and one clear prototype. Treat extra ideas as optional extensions with explicit MVP gate criteria before supporting goals begin.	Medium
FL workstream led by a single team member; illness or workload from other courses creates a bottleneck with no backup	Possible	Moderate	High	Maintain shared FL documentation in Notion. Cross-train at least one other member on the Flower framework during Phase 2 so a second team member can step in if needed.	Low

Undocumented design decisions in Part A make it difficult to resume or extend work at the start of Part B	Possible	Moderate	High	Document every design decision in Notion Blueprint pages before implementation begins. End Part A with a written handover note summarizing outstanding issues and the recommended starting point for Part B.	Low
System/Implementation Risks					
Simulated traffic may not reflect real C-V2X behavior closely enough	Possible	Moderate	High	Clearly define assumptions, justify the simulation setup, and discuss realism limits in the report. Validate generated dataset against 5G-NIDD feature distributions before downstream use.	Medium
IDS may show poor accuracy or high false positive rates under changing traffic conditions	Possible	Severe	Catastrophic	Compare multiple models, adjust classification thresholds, and adapt method incrementally. Report per-class false positive rate separately across all five attack types, not just aggregate accuracy.	Medium
Integration issues between datasets, simulator, IDS model and Federated Learning framework	Likely	Moderate	High	Build and test each component separately, then integrate in stages using version control. Define a common intermediate data format in Phase 1 before implementation begins.	Low
The simulation may become too slow or too complex when scaling up to many vehicles or nodes	Possible	Moderate	High	Start with a minimum three-node setup, then scale gradually only if time and computing resources allow. Use RMIT HPC cluster as primary compute before any AWS spend.	Medium
Sharing model updates may raise privacy concerns	Possible	Moderate	High	Avoid storing unnecessary raw data, document how updates are handled, and discuss privacy limits clearly in the report. Note gradient inversion as a known limitation of standard FedAvg federation.	Low
Poisoned model updates injected during the Byzantine experiment may contaminate final evaluation results if not correctly isolated	Unlikely	Moderate	Medium	Run Byzantine injection as a separate, clearly labelled experiment with independent model checkpoints. Never mix poisoned and clean run results in the same evaluation table.	Low
Personal Safety Risks					
Ergonomic risk from extended computer use during intensive development and simulation periods	Possible	Minor	Medium	Follow standard guidance on regular screen breaks and appropriate workstation setup. Distribute workload across team members to avoid concentrated periods of extended individual work.	Low
Electrical hazards from chargers or cables during standard computer use	Unlikely	Negligible	Low	Keep workspace tidy and secure cables. No laboratory or workshop access is required for this project, so no electrical induction or equipment risk assessment forms apply.	Low

6.2 Ethical Considerations

Ethical considerations are integrated throughout the project rather than treated as a separate requirement. Since the project focuses on a federated learning based IDS for C-V2X networks, privacy is a central ethical issue. The proposed design reduces unnecessary raw data centralization by keeping training local and sharing only model updates [2], which supports a more privacy-aware approach than traditional centralized data collection. Raw V2X traffic constitutes personal data under GDPR and centralizing it creates a high-value target for adversaries [7]. However, research has demonstrated that model weight updates shared during federated training can be used to reconstruct the original training data through gradient inversion, meaning the privacy benefit of federation should be treated as a reduction in risk rather than its complete elimination [20]. Ethical practice also includes using appropriate public or open datasets, documenting how data and model updates are handled, avoiding unnecessary retention of raw data, and being transparent about the limitations of simulation-based results. Because this is a capstone prototype rather than a deployed road-safety product, the report should avoid overstating real-world readiness and should discuss limitations honestly, especially around dataset realism, latency, scalability, and any potential false positives. In a safety-critical vehicular system, a false positive is more than a classification error; it can trigger evasive maneuvers such as emergency braking that create the same collision risk the system was designed to prevent, which is why false positive rate is included as a primary evaluation metric [8].

6.3 OHS Considerations

This project is conducted entirely through software development and simulation. The only occupational health and safety consideration is ergonomic risk from sustained computer use during intensive development periods, which is managed through regular screen breaks, appropriate workstation setup, and distribution of workload across team members.

6.4 AI Usage Declaration

Artificial intelligence tools were used in the preparation of this document in accordance with RMIT University's AI Usage Framework. An AI Declaration Form is submitted alongside this assessment task as required.

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Appendix

Appendix 1 – Risk Look-up Table Matrix

Risk Look-up Table		Impact			
		Negligible (little to no effect)	Minor (Noticeable effect, resolved easily)	Moderate (Impact may require review of outcomes)	Severe (could seriously impact outcomes)
Likelihood	Unlikely	Low	Medium	Medium	High
	Possible	Low	Medium	High	Catastrophic
	Likely	Medium	High	High	Catastrophic
	Highly likely	Medium	High	High	Catastrophic

Appendix 2 - Gantt chart

